

Space Shooter Game

Clone Repo : <https://github.com/codertheashish/Space-Shooter-Game>

1. Introduction

The **Space Shooter Game** is a 2D arcade shooting game developed using **Python** and the **Pygame** library. In this game, the player controls a spaceship at the bottom of the screen and shoots enemies falling from the top. The player has **5 lives**, and each collision with an enemy decreases one life. The game becomes more difficult because the enemy speed increases over time. When all lives are lost, a **Game Over** screen appears with options to restart or exit the game.

2. Purpose

The main purpose of this project is to learn **2D game development** using Python and Pygame while understanding game mechanics and object interaction.

This project helps in learning:

- Game loop
 - Player movement
 - Bullet shooting system
 - Enemy spawning
 - Collision detection
 - Score management
 - Lives system
 - Game Over handling
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3. Technologies Used

Technology	Purpose
Python	Main programming language
Pygame	Creates the game window, graphics, animations, and keyboard controls
Random Module	Generates random enemy positions and movement direction
Sys Module	Closes the game safely when the player exits

4. Main Features

✓ Player Movement

The player controls a spaceship placed at the bottom of the screen.

Movement keys:

- **Left Arrow / A** → Move Left
- **Right Arrow / D** → Move Right

The spaceship cannot move outside the game window.

✓ Shooting System

The player fires bullets by pressing the **Spacebar**.

- Every bullet moves upward.
- Bullets destroy enemies when they collide.
- A new bullet is created every time the Space key is pressed.

The project uses **larger bullets**, making it easier to hit enemies.

✓ Enemy System

The game starts with **6 enemies**.

- Enemies fall from the top of the screen.
- When an enemy is destroyed, a new one appears at a random position.
- If an enemy leaves the screen, another enemy is generated automatically.

This keeps the gameplay continuous without stopping.

✓ Dynamic Difficulty

The enemy speed slowly increases while the game is running.

As a result:

- The game becomes more challenging.
 - Players need faster reactions.
 - The difficulty increases naturally over time.
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✓ Lives System

The player starts with **5 lives**.

- Every collision between the spaceship and an enemy removes **1 life**.
 - The game continues until all lives are lost.
 - Remaining lives are displayed on the screen throughout the game.
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✓ Score System

The score is displayed in the top-left corner of the screen.

- The score increases whenever the player fires a bullet.
- The current score is visible during the entire game.

This gives the player continuous feedback while playing.

✓ Game Over Screen

When the player's lives reach **zero**, the game displays a Game Over screen.

The player can:

- **Press R** → Restart the game.
- **Press ESC** → Exit the game.

The final score is also shown on the Game Over screen.

✓ Animated Space Background

The game creates a simple animated space environment.

- Small stars are drawn randomly on the screen.
 - The black background gives the appearance of outer space.
 - This improves the visual experience without affecting performance.
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Summary

"This project is a 2D Space Shooter game developed using Python and the Pygame library. The player controls a spaceship, shoots enemies, and avoids collisions. The game includes a shooting system, enemy spawning, increasing difficulty, score tracking, a five-life system, and a Game Over screen with restart and exit options. Through this project, I learned object movement, collision detection, game loops, event handling, and real-time game development using Pygame."